

INFINITY FITNESS STUDIOS

PROJECT OVERVIEW & BUSINESS PROBLEM

Member Retention, Utilization & Operational Analytics
December 2025 - February 2026

Document Control	Details
Document ID	IFS-PO-001
Version	1.0
Role	Data Analyst
Engagement	Direct client engagement
Business	Gym / fitness studio
Data Coverage	Approximately one year of historical business data
Primary Data Source	Excel
Tools Confirmed	Python, SQL, Power BI
Confidentiality	Client-level datasets, commercially sensitive figures and identifiable operational details are excluded from the portfolio documentation.

1. Engagement Context

Infinity Fitness Studios had operational and membership data available, but management lacked consolidated analytical visibility into member retention, attrition, membership usage, renewals, utilization patterns and related operating performance. The engagement focused on converting the available historical data into decision-oriented analysis and Power BI reporting.

2. Business Problem

The central problem was not the absence of data. The studio had member, plan, payment, attendance, renewal, trainer, booking, revenue and expense information, but the data was not clean or organized for consistent analysis. This limited management's ability to understand when member engagement was declining, how membership plans were performing, when the facility was under- or highly utilized, and where pricing or operational interventions could be considered.

3. Confirmed Data Available

Data Area	Confirmed Information
Members	Member-level information
Membership	Membership plans; joining dates; renewal dates
Plans	Monthly, 3-month, 6-month, annual and personal-training offerings
Payments	Member payment information
Attendance	Attendance / check-in information
Trainers	Trainer information and member assignments
Classes / Sessions	Class bookings and trainer sessions
Financial	Revenue and expense information
Time Coverage	Approximately one year
Data Condition	Source data required cleaning before analysis

4. Analytical Objectives

- Measure and understand member retention and churn/attrition behavior.
- Track attendance and membership usage to identify utilization patterns.
- Analyze membership renewals and plan performance.
- Identify peak and underutilized periods using attendance by hour/day.
- Evaluate trainer utilization using sessions, bookings, workload, member assignments and attendance.
- Analyze revenue and related business performance where supported by the available data.
- Use cohort analysis and member-duration segmentation to identify changes in engagement over time.
- Translate findings into practical pricing and operational recommendations.

5. Analysis Approach

Stage	What Was Done
Requirements Discovery	Calls, WhatsApp discussions, meetings, review of existing reports and review of Excel data with the gym manager.
Data Preparation	Cleaned and prepared the source data for analysis. Exact row counts and detailed client data are intentionally not disclosed.
Exploratory Analysis	Used Python for cleaning, EDA, segmentation, calculations and visualization; SQL was also used during the analytical workflow.
Cohort Analysis	Grouped members by joining month and tracked attendance / retention across subsequent periods.
Member Segmentation	Analyzed behavior by membership duration and attendance patterns.
Utilization Analysis	Analyzed attendance by hour/day and membership usage; evening activity emerged as an important utilization period.
Trainer Analysis	Reviewed sessions, class bookings, workload, member assignments and attendance.
Dashboarding	Built the Power BI dashboard from scratch to consolidate retention, utilization and operating insights.
Business Communication	Delivered analysis through Power BI, Excel analysis, presentation, written recommendations, Python notebook, report and live walkthrough.

6. Key Finding: Post-New-Year Engagement Decline

Cohort and attendance analysis identified a decline in member engagement following the New Year period, with the decline becoming visible approximately one month later. The analysis was based on joining-month cohorts, subsequent attendance/retention behavior, membership-duration segmentation and comparison of member activity over time.

For portfolio purposes, this finding is described without confidential member counts, churn percentages or revenue values. It should not be simplified into a fixed claim that every member churned after three months.

7. Churn & Retention Interpretation

Concept	Project Interpretation
Attrition / Churn	Member disengagement reflected through stopped attendance and/or membership cancellation.
Retention	Percentage of members who continue their membership over the relevant analysis period.
Cohort View	Members grouped by joining month and followed through subsequent attendance / retention periods.
Renewal View	Membership renewal behavior analyzed alongside plan and attendance information.

8. Business Recommendations

- Use discounts around membership renewal where appropriate.
- Encourage longer-duration membership plans through targeted incentives.

- Use re-engagement offers for inactive members.
- Consider quarterly promotional offers.
- Use utilization insights to support operational decisions around busy and underutilized periods.
- Use trainer workload, bookings and attendance visibility to support staffing and trainer-allocation decisions.

9. Outcome & Business Value

The work provided management with clearer visibility into renewal behavior, member engagement and underutilized periods. The gym manager provided positive feedback, and recommendations were used in the business. Because exact post-implementation metrics are not retained or cannot be disclosed, this case study does not attach unsupported revenue, retention or utilization improvement percentages.

10. Deliverables

Deliverable	Confirmed
Power BI Dashboard	Yes - built from scratch
Excel Analysis	Yes
Presentation	Yes
Written Recommendations	Yes
Python Notebook	Yes
Report	Yes
Live Walkthrough	Yes

11. Portfolio Scope & Confidentiality

This case study documents the analytical approach, business problem, methods and high-level outcomes of a real client engagement. Client-level datasets, member records, commercially sensitive figures, detailed internal operating information and other restricted details are intentionally excluded. Where exact figures are unavailable or confidential, the project documentation uses qualitative descriptions rather than fabricated values.

12. Next Portfolio Artifacts

- Business Requirements Document (BRD)
- User Stories & Acceptance Criteria
- KPI Catalogue
- Data Dictionary
- Analysis & Segmentation Methodology
- Power BI Requirements
- Data Model & DAX Measures
- Dashboard evidence / sanitized portfolio visuals
- Findings & Recommendations
- Data Validation & Dashboard Testing

13. Interview-Ready Project Summary

As the Data Analyst for Infinity Fitness Studios, I worked with approximately one year of membership, attendance, payment, trainer, booking, revenue and expense data. I cleaned and analyzed the data using Python, SQL and Power BI, built the dashboard from scratch, and used cohort analysis and segmentation to understand retention, attrition and utilization. The analysis highlighted a post-New-Year decline in member engagement, improved visibility into renewal and underutilized

periods, and supported pricing and operational recommendations including renewal incentives, longer-duration plans and re-engagement offers.